Online Resource 2： README

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Discrimination of deposit and metal types based on stibnite trace elements: A machine learning method

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1 Purpose and files

This package contains the random forest (RF) metal-type classifier and linear support vector machine (SVM) deposit-type classifier used in the article. The scripts reproduce model evaluation and fit models for application to new stibnite analyses. Model settings, reference deposits and class labels are fixed.

File	Purpose
`rf_model.py`	Predict Sb, Au and Au–Sb metal types using Cu, Zn, As, Se, Ag, Sn and Pb
`svm_model.py`	Predict four deposit types using Te, Cu, Pb, Zn, As and Ge
`README.pdf`	Installation, input requirements, commands and result interpretation

2 Installation

The code was tested with Python 3.12.14.

```
python -m pip install numpy==2.3.5 pandas==2.2.3 scipy==1.17.0
python -m pip install scikit-learn==1.8.0 joblib==1.5.3
```

3 Input data

Use UTF-8 CSV files with one spot analysis per row and original concentrations in *ppm*. The scripts apply the *log10* transformation.

RF reference file (`rf_points.csv`). Include `Deposit`, Cu, Zn, As, Se, Ag, Sn, Pb, `Frozen_Row` and `Keep_after_score<=4`. Keep the original, unique and increasing `Frozen_Row` identifiers and row order;

SVM reference file (`svm_points.csv`). Include `Deposit`, Te, Cu, Pb, Zn, As and Ge for all 941 analyses from the 17 reference deposits.

New data (`new_points.csv`). Include the relevant element columns and a `Unit` column identifying each deposit or defined mineral population. Reference-only row identifiers and quality-control flags are unnecessary. If the grouping column is `Deposit`, add `--unit-col Deposit` to the prediction command.

4 Running the code

Extract the ZIP and open a terminal in the script directory. The examples assume that the CSV files are in that directory. Use file paths if they are stored elsewhere.

RF metal-type classification

```
python rf_model.py lodo --input rf_points.csv
python rf_model.py fit --input rf_points.csv
python rf_model.py predict --input new_points.csv --model rf_fitted.joblib
```

SVM deposit-type classification

```
python svm_model.py lodo --input svm_points.csv
python svm_model.py fit --input svm_points.csv
python svm_model.py predict --input new_points.csv --model svm_fitted.joblib
```

`lodo` runs leave-one-deposit-out cross-validation. `fit` trains on the full reference set and must precede `predict`. Predictions use the saved preprocessing and classifier. Add `--help` after a command to view its options.

5 Outputs and interpretation

RF. `Direct_RF` is the class with the highest mean spot probability. `Final_Metal_Type` applies the study's correction. The output reports both assignments and the percentile; mean probabilities remain unchanged.

SVM. The script represents each unit by the 75th percentile (P75) of log10-transformed positive concentrations. Output classes are G1, orogenic hydrothermal; G2, epithermal hydrothermal; G3, intrusion-related hydrothermal; and G4, basinal/meteoric fluid.